

Appendix R9: Classification Models

Binomial and Categorical Classification Models

J. Alberto Espinosa

2025-01-04

This Quarto document contains R code examples about classification models. A classification model predicts the correct classification of an outcome rather than its quantitative value. Models with only two possible outcomes are called **Binomial** and those with more than two outcomes are called **Multinomial**. The standard baseline classification model is the Binomial or Multinomial Logistic Regression, which I briefly introduced in Chapter 3. I also discuss other popular classification models, like Linear (LDA) and Quadratic Discriminant Analysis (QDA), and popular methods and metrics to evaluate model accuracy, such as the Confusion Matrix, Misclassification rates, Sensitivity and Specificity, Receiver Operating Characteristics (ROC) Curve, and Area under the ROC Curve (AUC). For the corresponding conceptual material associated with these appendices, consult the main book [Predictive Analytics and Machine Learning for Managers](#). Chat with the PAML4M book and its related lecture slides and R scripts with its companion [PAML4M ChatBot](#).

Table of contents

1. Introduction	2
Overview	2
The Generalized Linear Model Function glm()	3
2. Binomial Logistic Regression	3
Logistic Regression Fit Statistics	4
From Log-Odds to Odds	5
3. The Confusion Matrix	6
Cross-Validation Train and Test Sub-Samples	6
Confusion with cross-Validated Test Results	7
4. The Receiver Operating Characteristics (ROC) Curve	11
5. Multinomial Logistic Regression	13
6. Linear Discriminant Analysis (LDA)	17
ROC Curve and LDA	25
7. Quadratic Discriminant Analysis (QDA)	26

This script was created by J. Alberto Espinosa for educational and training purposes. Feel free to use this material for your own work, but please do not share or duplicate without the author's permission.

1. Introduction

Overview

Classification models predict outcomes that are either binary or categorical. There are several types of classification models, including Logistic regression, classification trees, linear and quadratic discriminant analysis (LDA and QDA), and classification neural networks, among others. We will be using the Stock Market **{ISLR}Smarket** data set frequently in this section. Before you continue, review the variables in this data set by typing **?Smarket** at the console.

```
library(ISLR) # Contains the Smarket data set
dim(Smarket) # Dimensions of the Smarket data set
```

```
[1] 1250    9
```

This data set does not yield very interesting results in Logistic models but the results illustrate a few important points. It also illustrates the use of lag transformations in logistic models. Let's explore the data:

```
names(Smarket)
```

```
[1] "Year"      "Lag1"      "Lag2"      "Lag3"      "Lag4"      "Lag5"
[7] "Volume"    "Today"     "Direction"
```

```
head(Smarket)
```

	Year	Lag1	Lag2	Lag3	Lag4	Lag5	Volume	Today	Direction
1	2001	0.381	-0.192	-2.624	-1.055	5.010	1.1913	0.959	Up
2	2001	0.959	0.381	-0.192	-2.624	-1.055	1.2965	1.032	Up
3	2001	1.032	0.959	0.381	-0.192	-2.624	1.4112	-0.623	Down
4	2001	-0.623	1.032	0.959	0.381	-0.192	1.2760	0.614	Up
5	2001	0.614	-0.623	1.032	0.959	0.381	1.2057	0.213	Up
6	2001	0.213	0.614	-0.623	1.032	0.959	1.3491	1.392	Up

One important thing to note is that the variable “Direction” is categorical, not binary. So, we have two choices, create a binary variable called “Up” = 1 when Direction is up and 0 otherwise. But in the example below we don't make conversions until later and we let R take care of this for us. Let's now explore the data:

```
summary(Smarket)
```

Year		Lag1		Lag2		Lag3	
Min.	:2001	Min.	:-4.922000	Min.	:-4.922000	Min.	:-4.922000
1st Qu.	:2002	1st Qu.	:-0.639500	1st Qu.	:-0.639500	1st Qu.	:-0.640000
Median	:2003	Median	: 0.039000	Median	: 0.039000	Median	: 0.038500
Mean	:2003	Mean	: 0.003834	Mean	: 0.003919	Mean	: 0.001716
3rd Qu.	:2004	3rd Qu.	: 0.596750	3rd Qu.	: 0.596750	3rd Qu.	: 0.596750
Max.	:2005	Max.	: 5.733000	Max.	: 5.733000	Max.	: 5.733000
Lag4		Lag5		Volume		Today	
Min.	:-4.922000	Min.	:-4.922000	Min.	:0.3561	Min.	:-4.922000
1st Qu.	:-0.640000	1st Qu.	:-0.640000	1st Qu.	:1.2574	1st Qu.	:-0.639500
Median	: 0.038500	Median	: 0.038500	Median	:1.4229	Median	: 0.038500
Mean	: 0.001636	Mean	: 0.00561	Mean	:1.4783	Mean	: 0.003138

```

3rd Qu.: 0.596750    3rd Qu.: 0.59700    3rd Qu.:1.6417    3rd Qu.: 0.596750
Max.    : 5.733000    Max.    : 5.73300    Max.    :3.1525    Max.    : 5.733000
Direction
Down:602
Up   :648

```

The Generalized Linear Model Function glm()

As discussed in Chapter 3, Ordinary Least Squares (OLS) regressions require the residuals to be normally distributed and the outcome variable to be continuous. When these assumptions don't hold, the Generalized Linear Model (GLM) can be used instead. OLS models are estimated with the `lm()` function in R, whereas GLM models are estimated with the `glm()` function. The syntax for the `glm()` function is the same as for the `lm()`, except that we need to provide two additional parameters:

- The **family distribution** of the residuals, specified with the `family =` parameter
- Any required **transformations** of the outcome variable to overcome the lack of continuity, specified with the `link =` function inside the family parameter.

A linear model with no transformation of the dependent variable estimated with `glm()` and a **gaussian** distribution yields the exact same results as **OLS** with the `lm()` function. But `glm()` also supports other distributions and transformations of the dependent variable besides the normal distribution.

Different regression methods require specific model `family =` and `link =` parameters in the `glm()` function:

- Binomial Logistic regression – `family = binomial(link = "logit")`
- OLS regression – `family = gaussian(link = "identity")` (default)
- Binomial Probit regression – `family = binomial(link = "probit")` - probit stands for **Probability Unit** and it works just like logit, except that it uses a different transformation function for Y.
- Count data models – `family = poisson(link = "log")`

2. Binomial Logistic Regression

In the next example, I will use the `Heart.csv` I had already downloaded from the ISLR website, but the actual data set and variable descriptions are available at: <https://rdrr.io/cran/ElemStatLearn/man/SAheart.html>. If you have not done so, download the data set and save it in CSV format as **Heart.csv** and then read it into a dataframe object named **heart**.

Let's read the data and take a look at a few records:

```

heart <- read.table("Heart.csv",
                    sep = ",",
                    head = T,
                    stringsAsFactors = T)

head(heart)

```

	sbp	tobacco	ldl	adiposity	famhist	typea	obesity	alcohol	age	chd
1	160	12.00	5.73	23.11	Present	49	25.30	97.20	52	1
2	144	0.01	4.41	28.61	Absent	55	28.87	2.06	63	1
3	118	0.08	3.48	32.28	Present	52	29.14	3.81	46	0
4	170	7.50	6.41	38.03	Present	51	31.99	24.26	58	1
5	134	13.60	3.50	27.78	Present	60	25.99	57.34	49	1
6	132	6.20	6.47	36.21	Present	62	30.77	14.14	45	0

Logistic model predicting coronary heart disease:

```
heart.fit <- glm(chd ~ .,
                family = binomial(link = "logit"),
                data = heart)
summary(heart.fit)
```

Call:

```
glm(formula = chd ~ ., family = binomial(link = "logit"), data = heart)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-6.1507209	1.3082600	-4.701	2.58e-06	***
sbp	0.0065040	0.0057304	1.135	0.256374	
tobacco	0.0793764	0.0266028	2.984	0.002847	**
ldl	0.1739239	0.0596617	2.915	0.003555	**
adiposity	0.0185866	0.0292894	0.635	0.525700	
famhistPresent	0.9253704	0.2278940	4.061	4.90e-05	***
typea	0.0395950	0.0123202	3.214	0.001310	**
obesity	-0.0629099	0.0442477	-1.422	0.155095	
alcohol	0.0001217	0.0044832	0.027	0.978350	
age	0.0452253	0.0121298	3.728	0.000193	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 596.11 on 461 degrees of freedom
Residual deviance: 472.14 on 452 degrees of freedom
AIC: 492.14

Number of Fisher Scoring iterations: 5

Looks like tobacco, ldl, family history, type a and age are the strongest predictors of coronary heart disease (chd = 1 yes; = 2 no). Interestingly Once you control for these factors obesity and alcohol are not significant predictors. Take a closer look:

Logistic Regression Fit Statistics

```
paste("2LL =", round(-2 * logLik(heart.fit), digits = 4))
```

```
[1] "2LL = 472.14"
```

```
paste("Deviance =", round(deviance(heart.fit), digits = 4)) # Same value
```

```
[1] "Deviance = 472.14"
```

There are various measures of deviance adjusted for model size, such as the Akaike Information Criterion (**AIC**), which is equal to $2LL + 2 * \text{Number of variables}$

```
AIC(heart.fit)
```

```
[1] 492.14
```

Let's display them all together

```
c("2LL" = -2 * logLik(heart.fit),  
  "Deviance" = deviance(heart.fit),  
  "AIC" = AIC(heart.fit))
```

	2LL	Deviance	AIC
	472.14	472.14	492.14

From Log-Odds to Odds

```
log.odds <- coef(heart.fit) # Get the coefficients  
odds <- exp(log.odds) # Convert log-odds to odds
```

We can also convert the odds to probabilities, but the probability effect of a coefficient is relative to the value of the predictors, so it changes as the predictors change, so they are not very useful for interpretations, other than saying that any coefficient with a probability greater than 0.5 increases the likelihood of a positive outcome, whereas less than 0.5 decreases the likelihood. Nevertheless, probabilities are very useful when predicting actual outcomes.

```
prob = odds / (1 + odds) # Convert odds to probabilities  
round(cbind("Log-Odds" = log.odds, # Display all 3 together  
          "Odds" = odds,  
          "Probabilities" = prob),  
      digits = 3)
```

	Log-Odds	Odds	Probabilities
(Intercept)	-6.151	0.002	0.002
sbp	0.007	1.007	0.502
tobacco	0.079	1.083	0.520
ldl	0.174	1.190	0.543
adiposity	0.019	1.019	0.505
famhistPresent	0.925	2.523	0.716
typea	0.040	1.040	0.510
obesity	-0.063	0.939	0.484
alcohol	0.000	1.000	0.500
age	0.045	1.046	0.511

95% Confidence intervals

With log-odds:

```
round(confint(heart.fit), digits = 3) # In log-odds
```

	2.5 %	97.5 %
(Intercept)	-8.776	-3.636
sbp	-0.005	0.018
tobacco	0.029	0.133
ldl	0.059	0.294
adiposity	-0.038	0.077
famhistPresent	0.481	1.376

typea	0.016	0.064
obesity	-0.152	0.022
alcohol	-0.009	0.009
age	0.022	0.069

With odds:

```
round(exp(confint(heart.fit)), digits = 3) # In odds
```

	2.5 %	97.5 %
(Intercept)	0.000	0.026
sbp	0.995	1.018
tobacco	1.029	1.142
ldl	1.061	1.342
adiposity	0.962	1.080
famhistPresent	1.618	3.958
typea	1.016	1.066
obesity	0.859	1.022
alcohol	0.991	1.009
age	1.022	1.072

3. The Confusion Matrix

Cross-Validation Train and Test Sub-Samples

Let's do some **Cross-Validation** testing with the **Confusion Matrix**.

Let's start by defining the train and test sub-samples using the **heart** data frame. First, let's set the default random number generator and random seed (to get repeatable results)

```
RNGkind(sample.kind = "default")
set.seed(1)
```

Then, let's create an index vector for the train sub-sample

```
train <- sample(1:nrow(heart),
               0.7 * nrow(heart))
```

We can now split the data into train and test sub-samples

```
heart.train <- heart[train, ] # Train sub-sample
nrow(heart.train) # Count the train observations
```

```
[1] 323
```

```
heart.test <- heart[-train, ] # Test sub-sample
nrow(heart.test) # Count the test observations
```

```
[1] 139
```

We can now train a Logistic regression model using the train sub-sample

```
heart.fit.train <- glm(chd ~ .,
                      family = binomial(link = "logit"),
                      data = heart.train)
```

We can now use the trained model to make predictions with the test data and evaluate the cross-validation accuracy of the model and build a confusion matrix with the results.

Confusion with cross-Validated Test Results

We use the `predict()` function to use the trained model `heart.fit.train` to make predictions with the test subset. We use the `type = "response"` attribute to get the probability of $Y = 1$.

```
heart.probs.test <- predict(heart.fit.train,
                           heart.test,
                           type = "response")
heart.probs.test[1:10] # Take a look at the first 10 predictions
```

	5	6	7	8	9	10	11
	0.74727697	0.64619084	0.19114356	0.62648581	0.13616443	0.58022506	0.59418432
	17	18	21				
	0.83491902	0.89829067	0.07528622				

To build the confusion matrix, we need to convert the predicted probabilities computed above into actual classifications of 0 (negative) or 1 (positive). This requires that we set a threshold **lambda** for the probability level to use to classify an observation either as a 1 or a 0. For this example, I am using **lambda** = 0.5, so that any observation with a predicted probability of being a positive (i.e., 1) greater than 0.5 or 50% is classified as a 1. We can alter this tuning parameter later.

```
lambda <- 0.5
```

Notice that I stored the threshold value in a variable named **lambda**. This way I can use the variable rather than the actual value 0.5 in the formulas below. This way, if I wish to change the threshold to another value, I just change it above and all the formulas that follow will work. You can try several values of **lambda** on your own.

Let's create a vector to store a binary value 1 if the probability of the outcome being 1 exceeds this threshold and 0 otherwise. Let's then display the probabilities of the outcome variable **chd** being 1 side by side with the actual classifications. All probabilities > 0.5 should have a 1, and a 0 otherwise.

```
heart.pred.test <- ifelse(heart.probs.test > lambda, 1, 0)
cbind("Prob chd = 1" = round(heart.probs.test, digits = 3),
      "Predicted Classification" = heart.pred.test)[1:10,]
```

	Prob chd = 1	Predicted Classification
5	0.747	1
6	0.646	1
7	0.191	0
8	0.626	1
9	0.136	0
10	0.580	1
11	0.594	1
17	0.835	1
18	0.898	1
21	0.075	0

Cross-Tabulation for the Confusion Matrix

Now we can construct the Confusion Matrix.

Technical Note: The construction of the confusion matrix that follows is a bit involved, but notice that once you construct the cross-tabulation with the results of any binomial classification model, if you name your cross-table **conf.mat**, the rest of the script lines will work without modification, so you will be able to use this script for any classification method you use with a simple copy/paste.

Let's use the `table()` function to cross-tabulate rows and columns of two vectors. The first vector has the classification predictions from the model using the test subset. The second vector has the actual classifications in the test subset.

```
conf.mat <- table("Predicted" = heart.pred.test,
                  "Actual" = heart.test$chd)
colnames(conf.mat) <- rownames(conf.mat) <- c("No", "Yes")
conf.mat
```

	Actual	
Predicted	No	Yes
No	74	26
Yes	16	23

We can easily add margin totals using the `addmargins()` function with 3 parameters: (1) the matrix object `conf.mat`; (2) `margin = 1` to display row totals, `margin = 2` to display column totals and `margin = c(1, 2)` to display row and column totals; and (3) `FUN = sum` to get margin totals.

```
conf.mat <- table("Predicted" = heart.pred.test,
                  "Actual" = heart.test$chd)
colnames(conf.mat) <- rownames(conf.mat) <- c("No", "Yes")

conf.mat <- addmargins(conf.mat,
                       margin = c(1, 2),
                       FUN = sum) # Add margin totals
```

Margins computed over dimensions
in the following order:
1: Predicted
2: Actual

```
conf.mat
```

	Actual			
Predicted	No	Yes	sum	
No	74	26	100	
Yes	16	23	39	
sum	90	49	139	

Technical note: `FUN` = can be `sum`, `mean`, `min`, `max`, etc. if you need other aggregations.

Once again, once you constructed **conf.mat** for whichever classification model you used, the rest that follows will be identical for any other binomial classification model.

Confusion Matrix Metrics

Let's compute true and false positives and negatives using variable names that are representative of the contents:


```

TruN <- conf.mat[1, 1] # True negatives
TruP <- conf.mat[2, 2] # True positives
FalN <- conf.mat[1, 2] # False negatives
FalP <- conf.mat[2, 1] # False positives

TotN <- TruN + FalP # Total actual negatives
TotP <- TruP + FalN # Total actual positives
TotNpr <- TruN + FalN # Total negative predictions
TotPpr <- TruP + FalP # Total positive predictions
Tot <- TotN + TotP # Total

```

If you computed margin totals above, you can just read them directly from the third row and column in the `conf.mat` matrix:

```

TotN <- conf.mat[3, 1]
TotP <- conf.mat[3, 2]
TotNpr <- conf.mat[1, 3]
TotPpr <- conf.mat[2, 3]
Tot <- conf.mat[3, 3]

```

Let's do a quick review of the results

```

cbind(TruN, TruP, FalN, FalP,
      TotN, TotP,
      TotNpr, TotPpr,
      Tot)

```

```

      TruN TruP FalN FalP TotN TotP TotNpr TotPpr Tot
[1,]   74   23   26   16   90   49   100    39 139

```

Confusion Matrix Accuracy and Error Rates

Aggregate rates

```

Accuracy.Rate <- (TruN + TruP) / Tot
Error.Rate <- (FalN + FalP) / Tot

```

Sensitivity or rate of true (i.e., correctly classified) positives

```

Sensitivity <- TruP / TotP

```

Specificity or rate of true (i.e., correctly classified) correct negatives

```

Specificity <- TruN / TotN

```

False Positive Rate = `1 - specificity`. This may seem unnecessary, but the rate of false positives is a popular accuracy metric and one often used in ROC curves. It is the ratio of observations classified positive, which were actually negative.

```

FalseP.Rate <- FalP / TotN # Or
FalseP.Rate <- 1 - Specificity # Same result

```

Let's combine all metrics in a vector (to compare with other models later)

```
logit.rates.50 <- c(Accuracy.Rate, Error.Rate,
                   Sensitivity, Specificity, FalseP.Rate)
names(logit.rates.50) <- c("Accuracy", "Error",
                          "Sensitivity", "Specificity", "False Pos")
print(logit.rates.50, digits = 2)
```

Accuracy	Error	Sensitivity	Specificity	False Pos
0.70	0.30	0.47	0.82	0.18

Changing the Classification Threshold

To illustrate how simple it is to change the threshold, I have copied all the instructions above in to the script lines that follow, except that I set the lambda threshold value to **0.6**. Two important technical issues about this: (1) if you use a different model and data set, all you need to do is construct the **conf.mat** confusion matrix and the rest of the script commands will work; (2) if you have experience programming, you can easily embed these script lines inside a loop function that can compute confusion matrices for several lambda values. You should try it.

```
lambda <- 0.6
heart.pred.test <- ifelse(heart.probs.test > lambda, 1, 0)
cbind("Prob chd = 1" = round(heart.probs.test, digits = 3),
      "Predicted Classification" = heart.pred.test)[1:10,]
```

	Prob chd = 1	Predicted Classification
5	0.747	1
6	0.646	1
7	0.191	0
8	0.626	1
9	0.136	0
10	0.580	0
11	0.594	0
17	0.835	1
18	0.898	1
21	0.075	0

Confusion Matrix with the threshold lambda = 0.6

```
conf.mat <- table("Predicted" = heart.pred.test,
                  "Actual" = heart.test$chd)
colnames(conf.mat) <- rownames(conf.mat) <- c("No", "Yes")

conf.mat <- addmargins(conf.mat,
                      margin = c(1, 2),
                      FUN = sum) # Add margin totals
```

Margins computed over dimensions
in the following order:
1: Predicted
2: Actual

```
conf.mat
```

		Actual		
Predicted	No	Yes	sum	
No	78	33	111	
Yes	12	16	28	
sum	90	49	139	

```

TruN  <- conf.mat[1, 1]
TruP  <- conf.mat[2, 2]
FalN  <- conf.mat[1, 2]
FalP  <- conf.mat[2, 1]

TotN  <- TruN + FalP
TotP  <- TruP + FalN
TotNpr <- TruN + FalN
TotPpr <- TruP + FalP
Tot    <- TotN + TotP

Accuracy.Rate <- (TruN + TruP) / Tot
Error.Rate   <- (FalN + FalP) / Tot
Sensitivity   <- TruP / TotP
Specificity   <- TruN / TotN
FalseP.Rate  <- 1 - Specificity

logit.rates.60 <- c(Accuracy.Rate, Error.Rate, Sensitivity,
                    Specificity, FalseP.Rate)
names(logit.rates.60) <- c("Accuracy", "Error", "Sensitivity",
                           "Specificity", "False Pos")

```

Compare the rates for both thresholds and see the sharp differences

```

logit.rates.compare <- rbind(logit.rates.50, logit.rates.60)

print(logit.rates.compare, digits = 2)

```

	Accuracy	Error	Sensitivity	Specificity	False Pos
logit.rates.50	0.70	0.30	0.47	0.82	0.18
logit.rates.60	0.68	0.32	0.33	0.87	0.13

4. The Receiver Operating Characteristics (ROC) Curve

To render ROC curves we use the **ROCR** library:

```
library(ROCR)
```

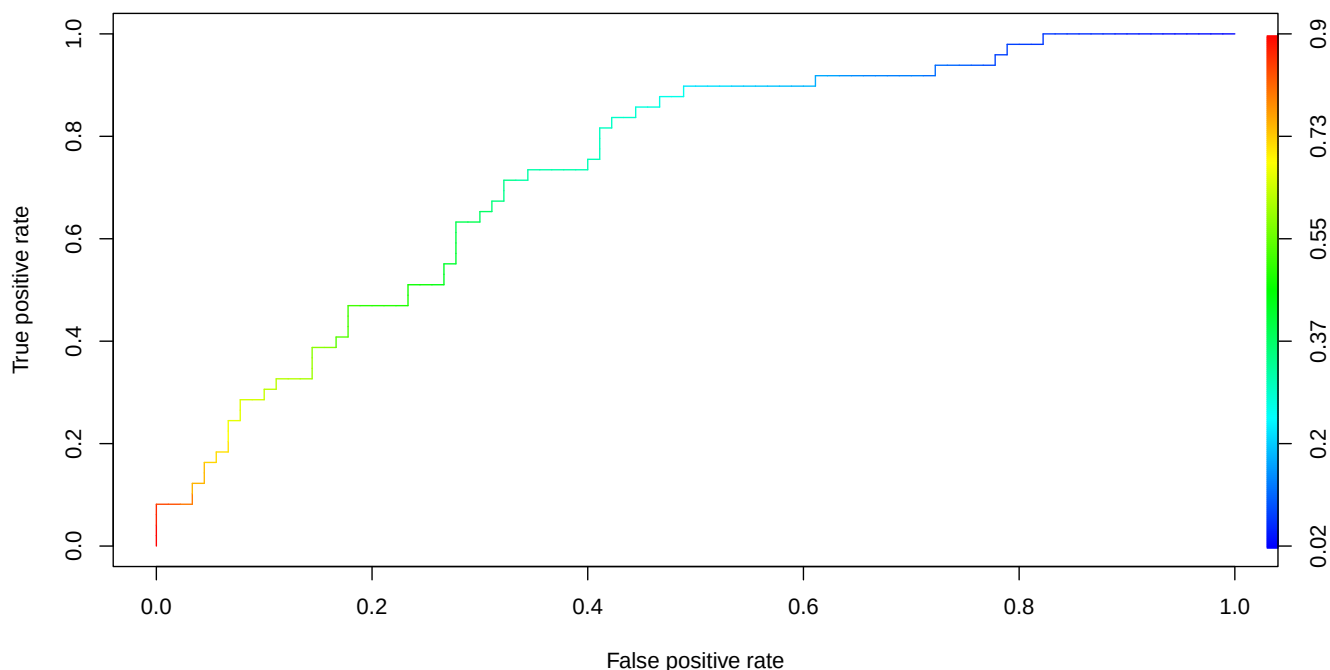
The first step with ROCR is to create a `prediction()` object. It is important to note that this function is contained in the `{ROCR}` library and it is **NOT** the same as the `predict()` function in the `{stat}` package. The `predict()` `{stat}` function is used to make predictions, whereas the `prediction()` `{ROCR}` function is used to render the ROC curve. The `prediction()` function needs two vectors as inputs:

- The predicted probabilities for the test sub-sample, as computed above in `heart.probs.test`, and
- The corresponding actual values of the outcome variable in the test sub-sample `heart.test$chd`

```
pred <- prediction(heart.probs.test, heart.test$chd)
```

The next step is to use the `performance()` {ROCR} function to create the ROC object. Use the parameter "tpr" for **True Positive Rate** in the vertical axis, and "fpr" for **False Positive Rate** in the horizontal axis. Other possible values are: "acc" for accuracy; "err" for error rate; "sens" for sensitivity; "spec" for specificity; and "auc" for area under the curve.

```
perf <- performance(pred, "tpr", "fpr")
plot(perf, colorize = T) # Plot the ROC curve with colors
```



Computing the AUC

```
auc <- performance(pred, "auc")
```

The `performance()` object above stores the **name** of the variable in `@y.name[[1]]` and the actual **AUC** in `@y.values[[1]]`. Note: the use of double brackets `[[ ]]` instead of single brackets `[ ]` and `@` instead of `$` to access values is because the `performance()` object is a **list** not a data frame. Lists use `[[ ]]` for indexing values and `@` for accessing elements in the list. Also, note that I rounded the AUC value to keep it short.

```
auc.name <- auc@y.name[[1]] # Retrieve the AUC name from the AUC list
auc.value <- round(auc@y.values[[1]], digits = 3) # Retrieve the AUC value
paste(auc.name, "is", auc.value) # Display them together
```

```
[1] "Area under the ROC curve is 0.739"
```

5. Multinomial Logistic Regression

A multinomial Logistic model is similar to a binomial Logistic model, except that the categorical outcome variable has more than 2 possible values. For example, suppose that you are trying to predict the likelihood that a real estate client will buy either a rural, suburban or urban home. In this case, the dependent variable is categorical and has 3 possible values. This model can be easily fitted using 2 binomial Logistic models, one for rural vs. suburban, and another for rural vs. urban. Notice that I indicated **rural** in both models, so that the log-odds effects measure the difference in effect between rural homes and each of the other two home locations. So, **rural** would be the **reference** value.

First, you need to select a baseline or **reference** value (e.g., rural) and then create dummy variables for the two other categories (i.e., suburban and urban). You then would need to build a binomial Logistic model with **suburban** = 1 for suburban homes, 0 if not, and then estimate the model. You could then do the same with **urban** = 1 for urban homes, 0 if not. Each regression of these binomial Logistic regressions will give you the effect on the log-odds of someone buying a suburban or rural house respectively.

This is a valid approach, but since you fit 2 separate models you will get two separate fit statistics for each of the models. In addition, there are other problems associated with splitting the data this way. The multinomial Logistic model estimates the two binomial models together as a single model (or more binomial models for multiple categories) and give you one set of fit statistics for the whole model.

More generally, a multinomial model is one in which the outcome variable has K possible categorical values. In our example above K = 3. More generally, a multinomial Logistic model fits logistic regression composed of K - 1 binomial Logistic models, but estimated together as a single model. There are R packages and functions that will estimate all K - 1 models jointly. Some popular ones are:

`multinom()`{`nnet`}, `mlogit`{`mlogit`}; and `vglm`{`VGAM`}. I use the `vglm()` (Vector Generalized Linear Model) function in the {`VGAM`} (Vector Generalized and Additive Models) package in the examples below. This function runs multinomial Logistic and other categorical outcome regressions.

The model is fitted with the `Womenlf` (it's women-el-f, not one-f) data set in the {`car`} packages, which contains women labor force statistics.

```
library(VGAM) # Contains the vglm() function
library(car)  # Contains the Womenlf (Women lf or Labor Force) data set
head(Womenlf) # Take a quick look at a few records
```

```
      partic hincome children  region
1 not.work      15  present Ontario
2 not.work      13  present Ontario
3 not.work      45  present Ontario
4 not.work      23  present Ontario
5 not.work      19  present Ontario
6 not.work       7  present Ontario
```

You can learn more about the `Womenlf` data set here:

```
? Womenlf # Use this command to explore the data set
```

Let's fit a model for labor participation (fulltime, not.work, parttime), based on husband's income in thousands and presence of children in the household (absence, presence). First, let's inspect the levels of the outcome variable `partic`.

```
levels(Womenlf$partic)
```

```
[1] "fulltime" "not.work" "parttime"
```

- The levels in \$partic are: fulltime, not.work, parttime
- fulltime is the refLevel (first alphabetically), but we can change this
- The effect for not.work will be suffixed :1 in the summary output
- The effect for parttime will be suffixed :2 in the summary output

```
vglm.fit <- vglm(partic ~ hincome + children,
                 family = multinomial(refLevel = 1), # "fulltime"
                 data = Womenlf)
summary(vglm.fit)
```

Call:

```
vglm(formula = partic ~ hincome + children, family = multinomial(refLevel = 1),
     data = Womenlf)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept):1	-1.98282	0.48418	-4.095	4.22e-05	***
(Intercept):2	-3.41513	0.66552	-5.132	2.87e-07	***
hincome:1	0.09723	0.02810	3.461	0.000539	***
hincome:2	0.10412	0.03328	3.128	0.001759	**
childrenpresent:1	2.55860	0.36220	7.064	1.62e-12	***
childrenpresent:2	2.58009	0.50972	5.062	4.15e-07	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1])

Residual deviance: 422.8819 on 520 degrees of freedom

Log-likelihood: -211.441 on 520 degrees of freedom

Number of Fisher scoring iterations: 5

No Hauck-Donner effect found in any of the estimates

Reference group is level 1 of the response

The summary output above is useful, but log-odds are difficult to interpret. It help to convert additive log-odds to multiplicative odds with the exp() function but we need to reconstruct the summary output, which can be done as follows:

Let's first save the summary() object:

```
vglm.sum <- summary(vglm.fit)
```

Then, extract coefficients and parameters from vglm.sum

```
round(
  cbind("Log Odds" = coef(vglm.fit), # or "Log Odds" = vglm.sum@coef3[,1]
        "Odds" = exp(coef(vglm.fit)), # or "Odds" = exp(vglm.sum@coef3[,1])
        "Std. Error" = vglm.sum@coef3[, 2],
        "z-Value" = vglm.sum@coef3[, 3],
        "p-Value" = vglm.sum@coef3[, 4]),
  digits = 4)
```

	Log Odds	Odds	Std. Error	z-Value	p-Value
(Intercept):1	-1.9828	0.1377	0.4842	-4.0952	0.0000
(Intercept):2	-3.4151	0.0329	0.6655	-5.1315	0.0000
hincome:1	0.0972	1.1021	0.0281	3.4607	0.0005
hincome:2	0.1041	1.1097	0.0333	3.1282	0.0018
childrenpresent:1	2.5586	12.9177	0.3622	7.0641	0.0000
childrenpresent:2	2.5801	13.1983	0.5097	5.0618	0.0000

Interpretations:

The coefficients in the multinomial Logistic regression, just like in binomial Logit, show the effect of a 1 unit increase in the predictor variable on the log-odds of the dependent variable. But this is the difference:

- In binomial Logistic, the coefficient of a predictor represents, on average and holding everything else constant, the **log-odds change** in units of the outcome variable being 1 (compared to 0), when the predictor increases by 1. The **odds** effect represents the factor by which the respective odds change.
- In multinomial Logistic, the coefficient is about the log-odds of the response variable being in that category, compared to the reference category (i.e., the response variable left out of the one specified with the `refLevel` = attribute)

Notice in the output above that each variable has 2 values. Since we have 3 categories in the response variable and the first one was left out as the reference variable, the first coefficient is for the log-odds of the second response category `not.work`; and the second one is for the third response category `parttime`.

`hincome:1` - on average and holding everything else constant, when the husband's income goes up by \$1K, the log-odds of `not working`, go up by 0.097, compared to working `full time`, The odds go up 1.102 times.

`hincome:2` - On average and holding everything else constant, when the husband's income goes up by \$1K, the log-odds of working `part time`, go up by 0.104. The odds go up 1.1097 times, compared to working `full time`.

`childrenpresent:1` - on average and holding everything else constant, when there are children present in the family, compared to no children, the log-odds of `not working`, go up by 2.559, compared to working `full time`, The odds go up 12.917 times.

`childrenpresent:2` - on average and holding everything else constant, when there are children present in the family, compared to no children, the log-odds of working `part time`, go up by 2.580, compared to working `full time`, The odds go up 13.198 times.

Fit Statistics

You can compute and display the deviance statistics as follows:

```
res.dev <- paste("2LL Residual Deviance",
                round(vglm.sum@criterion$deviance, digits = 2),
                "on", vglm.sum@df.residual, "degrees of freedom")
cat(res.dev, sep = "\n") # cat() with "\n" suppresses the annoying [1] index
```

2LL Residual Deviance 422.88 on 520 degrees of freedom

Or display the deviance statistics this way:

```
c("Log-Likelihood" = logLik(vglm.fit),
  "2LL" = -2 * logLik(vglm.fit),
  "Deviance" = deviance(vglm.fit),
  "AIC" = AIC(vglm.fit)) # Akaike Information Criterion
```

Log-Likelihood	2LL	Deviance	AIC
-211.4410	422.8819	422.8819	434.8819

Fit Statistics are similar to binomial Logistic with `glm()`. Notice that: the 2LL is 2 times the negative log-likelihood; 2LL and Deviance are the same thing; and AIC is like an adjusted 2LL (the more complex the model the larger the AIC, compared to the 2LL). As with binomial Logistic and other GLM models, the 2LL deviance is difficult to interpret by itself, but you can compute the Null deviance and evaluate the proportion of deviance explained by the model relative to the Null model. The `vglm()` function does not report the null deviance, but it can be easily computed by fitting a null model:

```
vglm.fit.null <- vglm(partic ~ 1,
                      family = multinomial(refLevel = 1),
                      data = Women1f)

null.dev <- round(deviance(vglm.fit.null), digits = 2)
model.dev <- round(deviance(vglm.fit), digits = 2)
dev.explained <- round((null.dev - model.dev) / null.dev, digits = 2)

c("Null Deviance" = null.dev,
  "Model Deviance" = model.dev,
  "Deviance Explained" = dev.explained) # Similar to R-squared
```

Null Deviance	Model Deviance	Deviance Explained
500.49	422.88	0.16

Predictions

Let's now find predicted values with the `predictvglm()` function (note: `newdata` = can be used to specify a test or new data set to predict if omitted or NULL the predictions are done on the training set).

```
pred.log.odds = predictvglm(vglm.fit,
                           newdata = NULL,
                           se.fit = T)
```

Just like with the coefficients, we get 2 predictions for each observation, one for the log-odds of being in the second category (compared to the first) and the other for the third category (compared to the first). Also, the `predictvglm()` object `pred.log.odds` has a few attributes.

```
log.odds <- pred.log.odds$fitted.values # Predicted values, log odds
se <- pred.log.odds$se.fit # Standard errors
pred.odds <- exp(pred.log.odds$fitted.values) # Predicted odds
pred.prob <- pred.odds/(1+pred.odds) # Predicted probabilities
```

Let's display all the results together:

```
pred.all <- round(
  cbind(log.odds,
        se,
        pred.odds,
        pred.prob),
  digits = 3)
colnames(pred.all) <- c("LogOdds2", "LogOdds3", # Need to add column names
                      "Std.Err2", "StdErr3",
                      "Odds2", "Odds3",
                      "Prob2", "Prob3")
```

Let's list the first 10 rows:


```
pred.all[1:10, ]
```

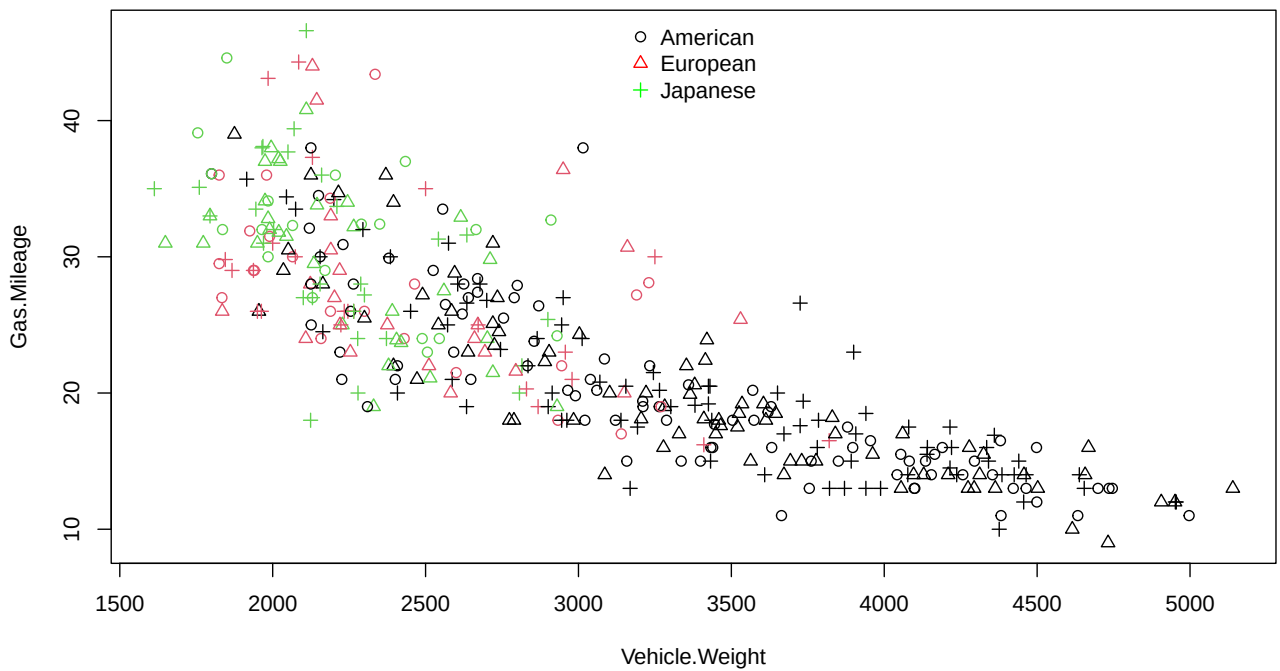
	LogOdds2	LogOdds3	Std.Err2	StdErr3	Odds2	Odds3	Prob2	Prob3
1	2.034	0.727	0.259	0.297	7.646	2.068	0.884	0.674
2	1.840	0.519	0.246	0.290	6.295	1.680	0.863	0.627
3	4.951	3.850	0.958	1.102	141.338	47.016	0.993	0.979
4	2.812	1.560	0.394	0.440	16.644	4.758	0.943	0.826
5	2.423	1.143	0.314	0.351	11.281	3.137	0.919	0.758
6	1.256	-0.106	0.282	0.353	3.513	0.899	0.778	0.473
7	2.034	0.727	0.259	0.297	7.646	2.068	0.884	0.674
8	1.256	-0.106	0.282	0.353	3.513	0.899	0.778	0.473
9	2.034	0.727	0.259	0.297	7.646	2.068	0.884	0.674
10	2.812	1.560	0.394	0.440	16.644	4.758	0.943	0.826

6. Linear Discriminant Analysis (LDA)

Let's take a look at a few attributes of the **Auto** data set in the **{ISLR}** library.

```
library(ISLR)
options(scipen = 4)

plot(Auto$mpg ~ Auto$weight,
     xlab = "Vehicle.Weight",
     ylab = "Gas.Mileage",
     col = Auto$origin,
     pch = c(1, 2, 3))
legend(x = "top",
      c("American", "European", "Japanese"),
      col = c("black", "red", "green"),
      pch = c(1, 2, 3),
      bty = "n")
```



The diagram above suggests that if we know the weight and gas mileage of a vehicle, we may be able to predict the likelihood that the vehicle is American, European or Japanese. Let's first try a simple Multinomial LDA model:

```
library(MASS) # Contains the lda() function
auto.fit.lda = lda(origin ~ weight + mpg,
                    data = Auto)
auto.fit.lda$prior # Check out the prior probabilities for each class
```

```
      1      2      3
0.6250000 0.1734694 0.2015306
```

```
auto.lda.pred <- predict(auto.fit.lda)
```

Let's look at the first 10 posterior probabilities for each class

```
round(auto.lda.pred$posterior[1:10, ], digits = 4)
```

```
      1      2      3
1  0.8964 0.0668 0.0367
2  0.9373 0.0434 0.0193
3  0.8855 0.0740 0.0405
4  0.9000 0.0679 0.0320
5  0.8953 0.0694 0.0353
6  0.9776 0.0153 0.0071
7  0.9796 0.0143 0.0061
8  0.9782 0.0153 0.0065
9  0.9818 0.0127 0.0055
10 0.9509 0.0339 0.0152
```

Let's now analyze a more complex data set with a binomial outcome. I use the **Heart.csv** data set provided by the ISLR package authors:

```
heart <- read.table("Heart.csv",
                    sep = ",",
                    head = T)

head(heart)
```

	sbp	tobacco	ldl	adiposity	famhist	typea	obesity	alcohol	age	chd
1	160	12.00	5.73	23.11	Present	49	25.30	97.20	52	1
2	144	0.01	4.41	28.61	Absent	55	28.87	2.06	63	1
3	118	0.08	3.48	32.28	Present	52	29.14	3.81	46	0
4	170	7.50	6.41	38.03	Present	51	31.99	24.26	58	1
5	134	13.60	3.50	27.78	Present	60	25.99	57.34	49	1
6	132	6.20	6.47	36.21	Present	62	30.77	14.14	45	0

Note: I used the same `Heart.csv` data file I used above for Logistic regression. I re-read here it for easy reference.

```
heart <- read.table("Heart.csv",
                    sep = ",",
                    head = T,
                    stringsAsFactors = T)

head(heart)
```

	sbp	tobacco	ldl	adiposity	famhist	typea	obesity	alcohol	age	chd
1	160	12.00	5.73	23.11	Present	49	25.30	97.20	52	1
2	144	0.01	4.41	28.61	Absent	55	28.87	2.06	63	1
3	118	0.08	3.48	32.28	Present	52	29.14	3.81	46	0
4	170	7.50	6.41	38.03	Present	51	31.99	24.26	58	1
5	134	13.60	3.50	27.78	Present	60	25.99	57.34	49	1
6	132	6.20	6.47	36.21	Present	62	30.77	14.14	45	0

Let's use cross-validation to evaluate the accuracy of the model, so we start by computing index vectors for the train and test sub-samples:

```
set.seed(1)
tr.size <- 0.7 # 70% train sub-sample
train <- sample(1:nrow(heart),
               tr.size * nrow(heart))
test <- seq(1:nrow(heart))[-train] # Any number not in the train vector
```

Let's now fit the model on the train data – i.e., train the LDA model

```
heart.fit.lda <- lda(chd ~ ., data = heart[train, ])
heart.fit.lda # Check out the results
```

Call:

```
lda(chd ~ ., data = heart[train, ])
```

Prior probabilities of groups:

```
0      1
0.6563467 0.3436533
```

Group means:

```
      sbp  tobacco      ldl adiposity famhistPresent      typea  obesity
0 134.9245 2.470283 4.373868 23.94382      0.2924528 51.98585 25.88882
1 142.9910 6.080450 5.625045 28.59550      0.6036036 53.74775 26.79865
      alcohol      age
0 15.43590 38.57075
1 19.71351 50.62162
```

Coefficients of linear discriminants:

```
          LD1
sbp      0.00232257478
tobacco  0.10180499306
ldl      0.16755841467
adiposity 0.01652028980
famhistPresent 0.84833975602
typea    0.02254757548
obesity  -0.05952142157
alcohol  0.00002954867
age      0.02994590316
```

Note: the `$scaling` attribute of this `lda()` object, `heart.fit.lda$scaling` contains the linear discriminant function coefficients displayed above. Note that these are NOT regression coefficients, but rather the equation of the linear discriminant function, which separates the outcome classifications into one category or another.

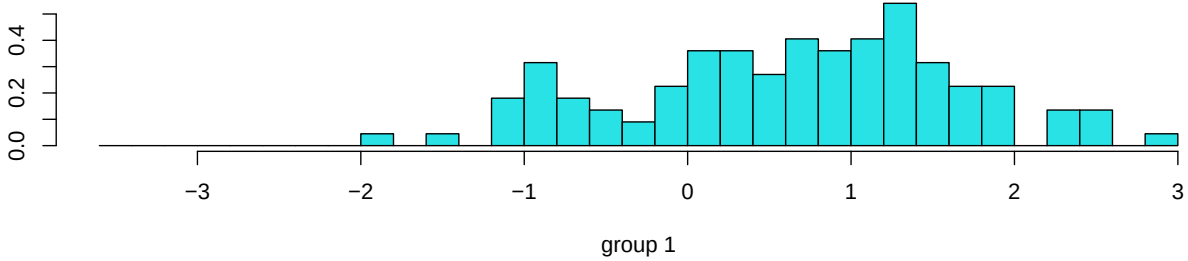
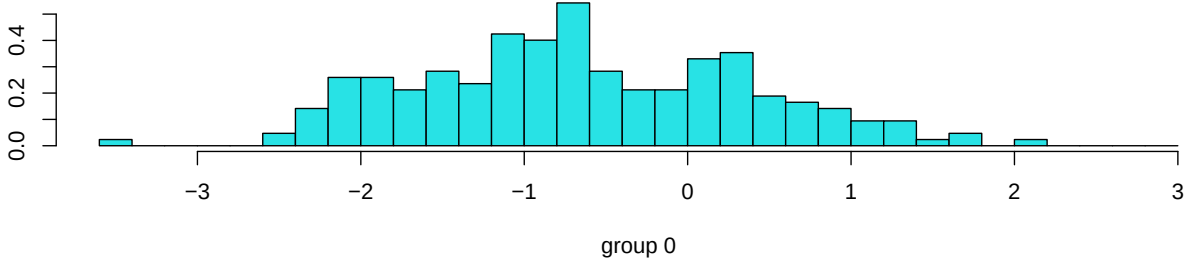
Another note: you can also use `*heart.fit.lda$prior` to get the prior probabilities or proportions of the response variable.

You can also get a summary, but it is not very useful

```
summary(heart.fit.lda)
```

```
      Length Class  Mode
prior      2      -none- numeric
counts     2      -none- numeric
means     18      -none- numeric
scaling     9      -none- numeric
lev        2      -none- character
svd         1      -none- numeric
N           1      -none- numeric
call       3      -none- call
terms      3      terms  call
xlevels    1      -none- list
```

```
plot(heart.fit.lda) # Outcome distribution for each outcom class
```



Let's now extract the test sub-sample

```
heart.test <- heart[test, ] # Test sub-sample
```

Now let's make predictions with the train model and the test sub-sample

```
heart.lda.pred <- predict(heart.fit.lda, heart.test)
heart.lda.pred$posterior[1:10, ] # Inspect the respective probabilities
```

	0	1
5	0.21907471	0.78092529
6	0.34229649	0.65770351
7	0.79750892	0.20249108
8	0.36714934	0.63285066
9	0.84219741	0.15780259
10	0.46317643	0.53682357
11	0.47794534	0.52205466
17	0.13991056	0.86008944
18	0.09529883	0.90470117
21	0.92396431	0.07603569

```
heart.lda.pred$class # Inspect the predicted classifications for prob > 0.5
```

```
[1] 1 1 0 1 0 1 1 1 1 0 0 0 0 0 1 0 0 0 0 0 0 1 1 0 0 0 1 0 0 0 0 0 0 0 1 1 1
[38] 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 1 1 0 0 0 0 0 0 0 1 1 1 0 0 1 0 0
[75] 0 1 0 1 0 0 0 0 1 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 1
[112] 0 0 0 1 0 0 1 0 0 1 0 0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Levels: 0 1
```

Note: the display above shows the prob of `chd = 0` and of `chd = 1`. That's why we see 2 columns. The default classification threshold is `prob > 0.5`. That is, any observation with a `prob(chd = 1 > 0.5)` will be predicted as 1, 0 otherwise. For most of what we do below, we only need the second column for `prob(chd = 1 > 0.5)`:

```
heart.lda.pred$posterior[1:10, 2] # Take a look
```

```
      5      6      7      8      9     10     11
0.78092529 0.65770351 0.20249108 0.63285066 0.15780259 0.53682357 0.52205466
      17     18     21
0.86008944 0.90470117 0.07603569
```

LDA Cross-Validated Confusion Matrix

Now that we have the trained model and some prediction objects, let's use them to construct the confusion matrix.

```
conf.mat <- table("Predicted" = heart.lda.pred$class,
                  "Actual" = heart.test$chd)
conf.mat
```

```
      Actual
Predicted 0  1
0      75 29
1      15 20
```

As we did before, let's compute accuracy the metrics and then add column and row totals and labels (as I mentioned earlier, if we name the cross-table `conf.mat`, then all the scripts we used earlier for the Logistic regression confusion matrix will work here without change)

```
TruN <- conf.mat[1, 1]
TruP <- conf.mat[2, 2]
FalN <- conf.mat[1, 2]
FalP <- conf.mat[2, 1]

TotN <- TruN + FalP
TotP <- TruP + FalN
TotNpr <- TruN + FalN
TotPpr <- TruP + FalP
Tot <- TotN + TotP
```

Quick check of the computations

```
cbind(TruN, TruP, FalN, FalP,
      TotN, TotP,
      TotNpr, TotPpr,
      Tot)
```

```
      TruN TruP FalN FalP TotN TotP TotNpr TotPpr Tot
[1,]   75   20   29   15   90   49   104   35 139
```

Let's add totals and labels and display the confusion matrix. In the Confusion Matrix example above we used the `addmargins()` function, which works here too, but I'm adding the margin totals manually below just to illustrate another method.

```

conf.mat.totals <- cbind(conf.mat,
                        c(TotNpr, TotPpr))
conf.mat.totals <- rbind(conf.mat.totals,
                        c(TotN, TotP, Tot))
colnames(conf.mat.totals) <-
  rownames(conf.mat.totals) <-
    c("No", "Yes", "Total")
conf.mat.totals # Display the confusion matrix

```

	No	Yes	Total
No	75	29	104
Yes	15	20	35
Total	90	49	139

Now let's use these metrics to compute accuracy and error rates

```

Accuracy.Rate <- (TruN + TruP) / Tot
Error.Rate <- (FalN + FalP) / Tot
Sensitivity <- TruP / TotP # Proportion of correct positives
Specificity <- TruN / TotN # Proportion of correct negatives
FalP.Rate <- 1 - Specificity # Proportion of false positives

```

Let's combine them into one vector and display the results

```

lda.rates.50 <- c(Accuracy.Rate, Error.Rate,
                  Sensitivity, Specificity, FalP.Rate)
names(lda.rates.50) <- c("Accuracy", "Rate", # Add column names
                        "Sensitivity", "Specificity", "False Pos")
lda.rates.50

```

Accuracy	Rate	Sensitivity	Specificity	False Pos
0.6834532	0.3165468	0.4081633	0.8333333	0.1666667

We can do the same with other values of lambda, say 0.60. We can use the data in the probability outcome vector we computed earlier to change the classification threshold to, say `prob(chd = 1 > 0.60)`. Notice that we use the `heart.lda.pred$posterior` vector we computed before. The probability outcomes did not change. We are only changing the threshold for when to classify an outcome as a 0 or a 1.

```

lambda <- 0.6
heart.lda.class.60 <- ifelse(heart.lda.pred$posterior[, 2] > lambda,
                             1, 0)
heart.lda.class.60 [1:20] # Take a look at a few revised classifications

```

5	6	7	8	9	10	11	17	18	21	30	32	34	46	47	52	54	55	57	58
1	1	0	1	0	0	0	1	1	0	0	0	0	0	1	0	0	0	0	0

Now, let's build the confusion matrix

```

conf.mat <- table(heart.lda.class.60, heart.test$chd)

TruN <- conf.mat[1, 1]
TruP <- conf.mat[2, 2]

```

```

FalN <- conf.mat[1, 2]
FalP <- conf.mat[2, 1]

TotN <- TruN + FalP
TotP <- TruP + FalN
TotNpr <- TruN + FalN
TotPpr <- TruP + FalP
Tot <- TotN + TotP

cbind(TruN, TruP, FalN, FalP, TotN, TotP, TotNpr, TotPpr, Tot)

```

```

      TruN TruP FalN FalP TotN TotP TotNpr TotPpr Tot
[1,]   77  17  32  13  90  49   109   30 139

```

Let's add totals and labels and display the confusion matrix

```

conf.mat.totals <- cbind(conf.mat,
                        c(TotNpr, TotPpr))
conf.mat.totals <- rbind(conf.mat.totals,
                        c(TotN, TotP, Tot))
colnames(conf.mat.totals) <-
  rownames(conf.mat.totals) <-
  c("No", "Yes", "Total")
conf.mat.totals # Display the confusion matrix

```

```

      No Yes Total
No    77  32  109
Yes   13  17   30
Total 90  49  139

```

Now, the accuracy statistics:

```

Accuracy.Rate <- (TruN + TruP) / Tot
Error.Rate <- (FalN + FalP) / Tot
Sensitivity <- TruP / TotP # Proportion of correct positives
Specificity <- TruN / TotN # Proportion of correct negatives
FalP.Rate <- 1 - Specificity # False Positive Rate

lda.rates.60 <- c(Accuracy.Rate, Error.Rate,
                  Sensitivity, Specificity, FalP.Rate)
names(lda.rates.60) <- c("Accuracy", "Error",
                        "Sensitivity", "Specificity", "False Pos")
round(lda.rates.60, digits = 3)

```

```

Accuracy      Error Sensitivity Specificity  False Pos
    0.676      0.324      0.347      0.856      0.144

```

Now, let's display the accuracy statistics with both thresholds together, so that we can compare results.

```

round(rbind(lda.rates.50, lda.rates.60), digits = 3)

```


	Accuracy	Rate	Sensitivity	Specificity	False Pos
lda.rates.50	0.683	0.317	0.408	0.833	0.167
lda.rates.60	0.676	0.324	0.347	0.856	0.144

Let's include the Binary Logistic accuracy metrics

```
round(rbind(logit.rates.50, logit.rates.60,
            lda.rates.50, lda.rates.60),
      digits = 3)
```

	Accuracy	Error	Sensitivity	Specificity	False Pos
logit.rates.50	0.698	0.302	0.469	0.822	0.178
logit.rates.60	0.676	0.324	0.327	0.867	0.133
lda.rates.50	0.683	0.317	0.408	0.833	0.167
lda.rates.60	0.676	0.324	0.347	0.856	0.144

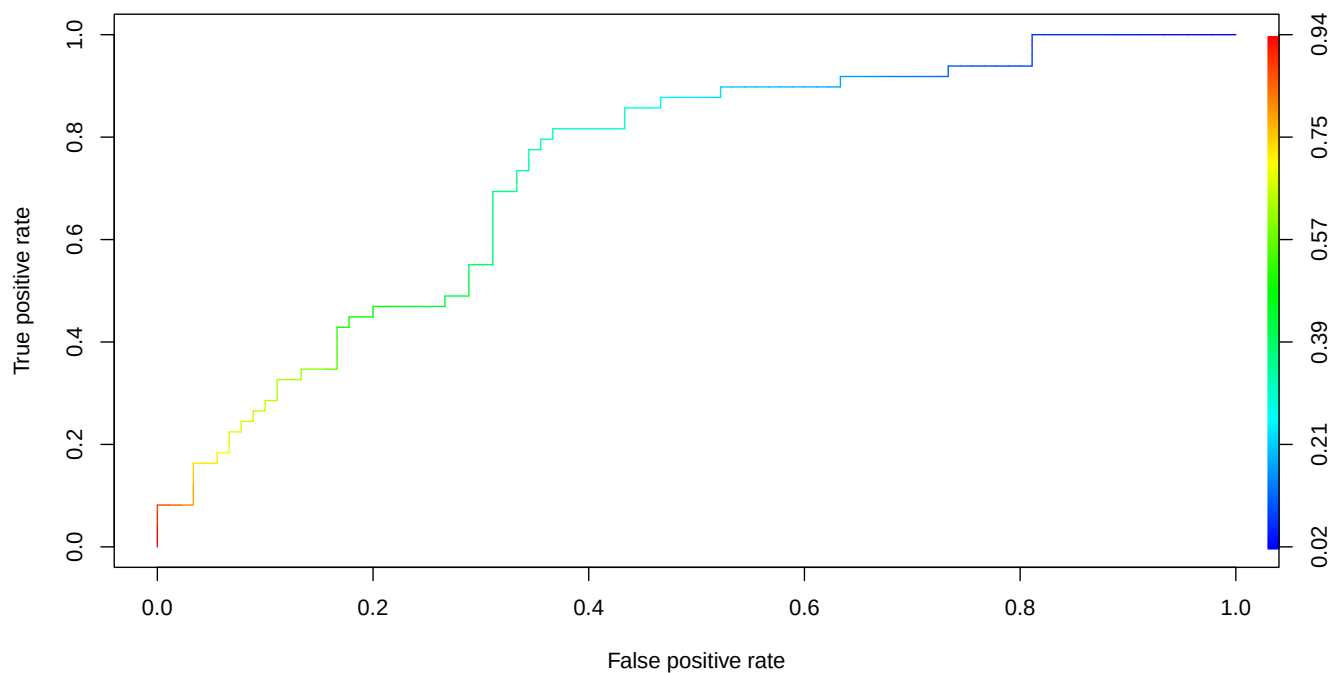
Which method is best overall? Which method is better for predicting positives? And for negatives? I think you have all the tools, information and wisdom to answer these questions now.

ROC Curve and LDA

We can construct the ROC curve for the LDA model in the exact same way we did for the Logistic regression model.

```
library(ROCR)
pred <- prediction(heart.lda.pred$posterior[, 2],
                  heart$chd[test])
perf <- performance(pred, "tpr", "fpr")

plot(perf, colorize=TRUE)
```



```
auc <- performance(pred,"auc") # Compute the AUC
c(auc@y.name[[1]], auc@y.values[[1]]) # Display the AUC
```

```
[1] "Area under the ROC curve" "0.734013605442177"
```

7. Quadratic Discriminant Analysis (QDA)

The syntax for **QDA** is identical to **LDA**, except that we use the `qda()` function instead of `lda()`. Both functions are in the `{MASS}` library. LDA and QDA fulfill similar goals and work in similar ways, except that LDA uses linear discriminant functions, whereas QDA uses quadratic discriminant functions. Which one is better? It all depends on the data and how it is distributed across the predictors. The best thing to do is to try both and evaluate them with cross-validation testing, as illustrated below.

```
library(MASS) # Contains the qda(){MASS} function
heart <- read.table("Heart.csv",
  sep = ",",
  head = T)
```

Compute index vectors for train and test sub-samples, as shown for LDA above. Then fit the QDA model using the `{MASS}qda()` function.

```
heart.fit.qda <- qda(chd ~ .,
  data = heart[train,])
```

Inspect the results

```
heart.fit.qda # See the linear discriminant function
```

Call:

```
qda(chd ~ ., data = heart[train, ])
```

Prior probabilities of groups:

```
      0      1
0.6563467 0.3436533
```

Group means:

```
      sbp  tobacco      ldl adiposity famhistPresent      typea  obesity
0 134.9245 2.470283 4.373868 23.94382      0.2924528 51.98585 25.88882
1 142.9910 6.080450 5.625045 28.59550      0.6036036 53.74775 26.79865
      alcohol      age
0 15.43590 38.57075
1 19.71351 50.62162
```

```
summary(heart.fit.qda) # Not very useful
```

```
      Length Class  Mode
prior      2    -none- numeric
counts     2    -none- numeric
means     18    -none- numeric
scaling  162    -none- numeric
ldet       2    -none- numeric
lev        2    -none- character
N          1    -none- numeric
call       3    -none- call
terms      3    terms  call
xlevels    1    -none- list
```

Let's extract the test sub-sample and make probability predictions

```
heart.test <- heart[test, ]
heart.qda.pred <- predict(heart.fit.qda, heart.test)
```

Now, let's take a look at some of the attributes of the **heart.qda.pred** object

```
heart.qda.pred$class # Inspect the classifications
```

```
[1] 1 1 0 0 0 1 1 1 1 0 0 0 0 0 1 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 1 0 1 1 1
[38] 0 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0 0 1 1 0 1 0 0 0 0 1 1 0 1 0 0 0 0 0
[75] 0 1 0 1 0 1 1 0 1 0 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 1
[112] 0 0 0 1 0 0 1 0 0 1 0 1 1 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Levels: 0 1
```

Check out the first 10 records of the posterior probabilities > 0.5 that **chd** is 0 and 1. Or just the probabilities for **chd = 1** (second column only)

```
heart.qda.pred$posterior[1:10, ] # Both probabilities
```

	0	1
5	0.06173275	0.938267248
6	0.31774112	0.682258877
7	0.92202810	0.077971899
8	0.82553199	0.174468010
9	0.98542350	0.014576499
10	0.29010239	0.709897612
11	0.18939482	0.810605179
17	0.01297368	0.987026318
18	0.06682875	0.933171250
21	0.99428720	0.005712804

```
heart.qda.pred$posterior[1:10, 2] # Just prob(chd = 1)
```

	5	6	7	8	9	10
0.938267248	0.682258877	0.077971899	0.174468010	0.014576499	0.709897612	
	11	17	18	21		
0.810605179	0.987026318	0.933171250	0.005712804			

Notice how any prediction probability of 0.5 for chd = 1 got a predicted classification of 1, and 0 otherwise.

Now let's work on the confusion matrix

```
conf.mat <- table("Predicted" = heart.qda.pred$class,
                  "Actual" = heart$chd[test])

TruN <- conf.mat[1, 1]
TruP <- conf.mat[2, 2]
FalN <- conf.mat[1, 2]
FalP <- conf.mat[2, 1]

TotN <- TruN + FalP
TotP <- TruP + FalN
TotNpr <- TruN + FalN
TotPpr <- TruP + FalP
Tot <- TotN + TotP

cbind(TruN, TruP, FalN, FalP, TotN, TotP, TotNpr, TotPpr, Tot)
```

	TruN	TruP	FalN	FalP	TotN	TotP	TotNpr	TotPpr	Tot
[1,]	70	21	28	20	90	49	98	41	139

Let's add totals and labels and display the confusion matrix

```
conf.mat.totals <- cbind(conf.mat,
                         c(TotNpr, TotPpr))
conf.mat.totals <- rbind(conf.mat.totals,
                         c(TotN, TotP, Tot))
colnames(conf.mat.totals) <-
  rownames(conf.mat.totals) <-
  c("No", "Yes", "Total")
conf.mat.totals # Display the confusion matrix
```

	No	Yes	Total
No	70	28	98
Yes	20	21	41
Total	90	49	139

Now, the accuracy statistics:

```

Accuracy.Rate <- (TruN + TruP) / Tot
Error.Rate <- (FalN + FalP) / Tot
Sensitivity <- TruP / TotP # Proportion of correct positives
Specificity <- TruN / TotN # Proportion of correct negatives
FalP.Rate <- 1 - Specificity # False Positive Rate

qda.rates.50 <- c(Accuracy.Rate, Error.Rate,
                 Sensitivity, Specificity, FalP.Rate)
names(qda.rates.50) <- c("Accuracy Rate", "Error Rate",
                        "Sensitivity", "Specificity", "False Positives")
round(rbind(qda.rates.50), digits = 3)

```

	Accuracy Rate	Error Rate	Sensitivity	Specificity	False Positives
qda.rates.50	0.655	0.345	0.429	0.778	0.222

As we did with LDA, we can use the data in the QDA probability outcome vector we computed earlier to change the classification threshold to, say `prob(chd = 1 > 0.60)`.

```

lambda <- 0.6
heart.qda.class.60 <- ifelse(heart.qda.pred$posterior[, 2] > lambda,
                             1, 0)
conf.mat <- table(heart.qda.class.60, heart.test$chd)

TruN <- conf.mat[1, 1]
TruP <- conf.mat[2, 2]
FalN <- conf.mat[1, 2]
FalP <- conf.mat[2, 1]

TotN <- TruN + FalP
TotP <- TruP + FalN
TotNpr <- TruN + FalN
TotPpr <- TruP + FalP
Tot <- TotN + TotP

cbind(TruN, TruP, FalN, FalP, TotN, TotP, TotNpr, TotPpr, Tot)

```

	TruN	TruP	FalN	FalP	TotN	TotP	TotNpr	TotPpr	Tot
[1,]	74	20	29	16	90	49	103	36	139

Let's add totals and labels and display the confusion matrix

```

conf.mat.totals <- cbind(conf.mat,
                         c(TotNpr, TotPpr))
conf.mat.totals <- rbind(conf.mat.totals,
                         c(TotN, TotP, Tot))
colnames(conf.mat.totals) <-
  rownames(conf.mat.totals) <-

```

```
c("No", "Yes", "Total")
conf.mat.totals # Display the confusion matrix
```

	No	Yes	Total
No	74	29	103
Yes	16	20	36
Total	90	49	139

Now, the accuracy statistics:

```
Accuracy.Rate <- (TruN + TruP) / Tot
Error.Rate <- (FalN + FalP) / Tot
Sensitivity <- TruP / TotP # Proportion of correct positives
Specificity <- TruN / TotN # Proportion of correct negatives
FalP.Rate <- 1 - Specificity # False Positive Rate

qda.rates.60 <- c(Accuracy.Rate, Error.Rate,
                  Sensitivity, Specificity, FalP.Rate)
names(qda.rates.60) <- c("Accuracy", "Error",
                        "Sensitivity", "Specificity", "False Pos")
round(qda.rates.60, digits = 3)
```

Accuracy	Error	Sensitivity	Specificity	False Pos
0.676	0.324	0.408	0.822	0.178

Now, let's display the accuracy statistics for borh, LDA and QDA, with both thresholds together, 0.5 and 0.6, so that we can compare results.

Let's include the Binary Logistic accuracy metrics

```
round(rbind(logit.rates.50, logit.rates.60,
            lda.rates.50, lda.rates.60,
            qda.rates.50, qda.rates.60),
      digits = 3)
```

	Accuracy	Error	Sensitivity	Specificity	False Pos
logit.rates.50	0.698	0.302	0.469	0.822	0.178
logit.rates.60	0.676	0.324	0.327	0.867	0.133
lda.rates.50	0.683	0.317	0.408	0.833	0.167
lda.rates.60	0.676	0.324	0.347	0.856	0.144
qda.rates.50	0.655	0.345	0.429	0.778	0.222
qda.rates.60	0.676	0.324	0.408	0.822	0.178

Which method is best overall? Which method is better for predicting positives? And for negatives? I think you have all the tools, information and wisdom to answer these questions now.